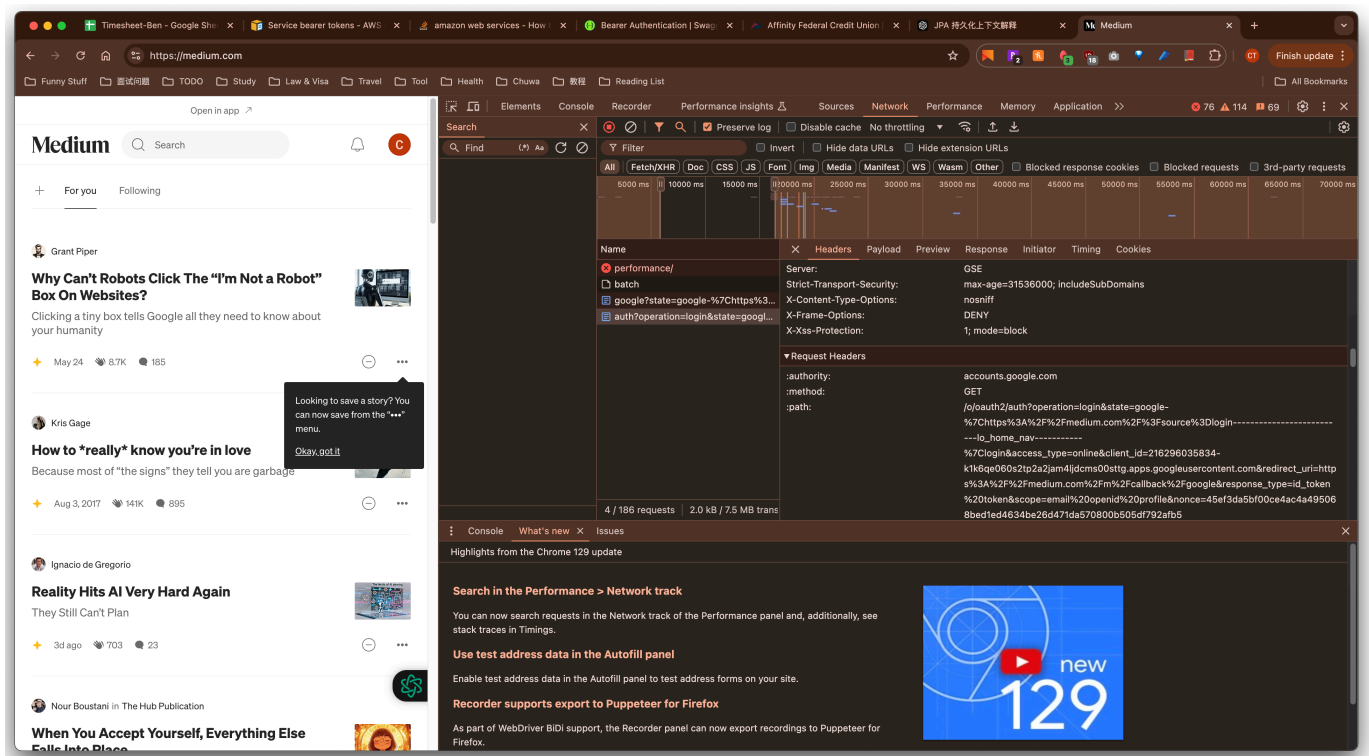


1. List all of the annotations you learned from class and homework to annotaitons.md
2. Explain **TLS**, **PKI**, **certificate**, **public key**, **private key**, and **signature**.
3. Write a Spring security based application, which provides **https** APIs (one simple get controller with empty response is good enough)instead of http, please generate a **self-signed** certificate to make your **https** TLS verification work.
 1. Pack your self-signed certificate in the form of **jks** file, as part of your application, name it properly
 2. Test if you can verify your HTTPs api without importing the self-signed certificate to your local certificate chain, if not, explain why.
 3. Explain what did you do to make **https** call work, **do NOT bypass TLS/SSL verfication in Postman (this is cheating)!**

Tutorial: <https://www.baeldung.com/spring-channel-security-https>
4. list all http status codes that related to authentication and authorization failures.
5. Compare **authentication** and **authorization**? Name and explain important components in Spring security that undertake **authentication** and **authorization**
6. Explain HTTP Session?
7. Explain Cookie?
8. Compare Session and Cookie?
9. Find **at least TWO** websites who can be logged in using your **Google Account**, explain in detail on how Google SSO works with screenshots like below, find SSO-related Rest calls in Chrome developer tool:



10. How do we use session and cookie to keep user information across the application?
11. What is the **spring security filter**?
12. Explain **bearer token** and how **JWT** works.
13. Explain how do we store sensitive user information such as password and credit card number in DB?
14. Compare UserDetails, AuthenticationProvider, AuthenticationManager, AuthenticationFilter?(把这几个名字看熟悉也行)
15. What is the disadvantage of Session? how to overcome the disadvantage?
16. how to get value from application.properties in Spring security?
17. What is the role of configure(HttpSecurity http) and configure(AuthenticationManagerBuilder auth)?
18. Reading, 泛读一下即可, 自己觉得是重点的, 可以多看看两眼。 <https://www.interviewbit.com/spring-security-interview-questions/#is-security-a-cross-cutting-concern>
 1. 1-12
 2. 17 - 30
19. Explain best practices to securely store secrets in applications.